

Hybrid DNS

Route 53 Resolver makes hybrid cloud easier for enterprise customers by enabling seamless DNS query resolution across your entire hybrid cloud. You can create DNS endpoints and conditional forwarding rules to allow resolution of DNS namespaces between your on-premises data center and AWS VPCs.

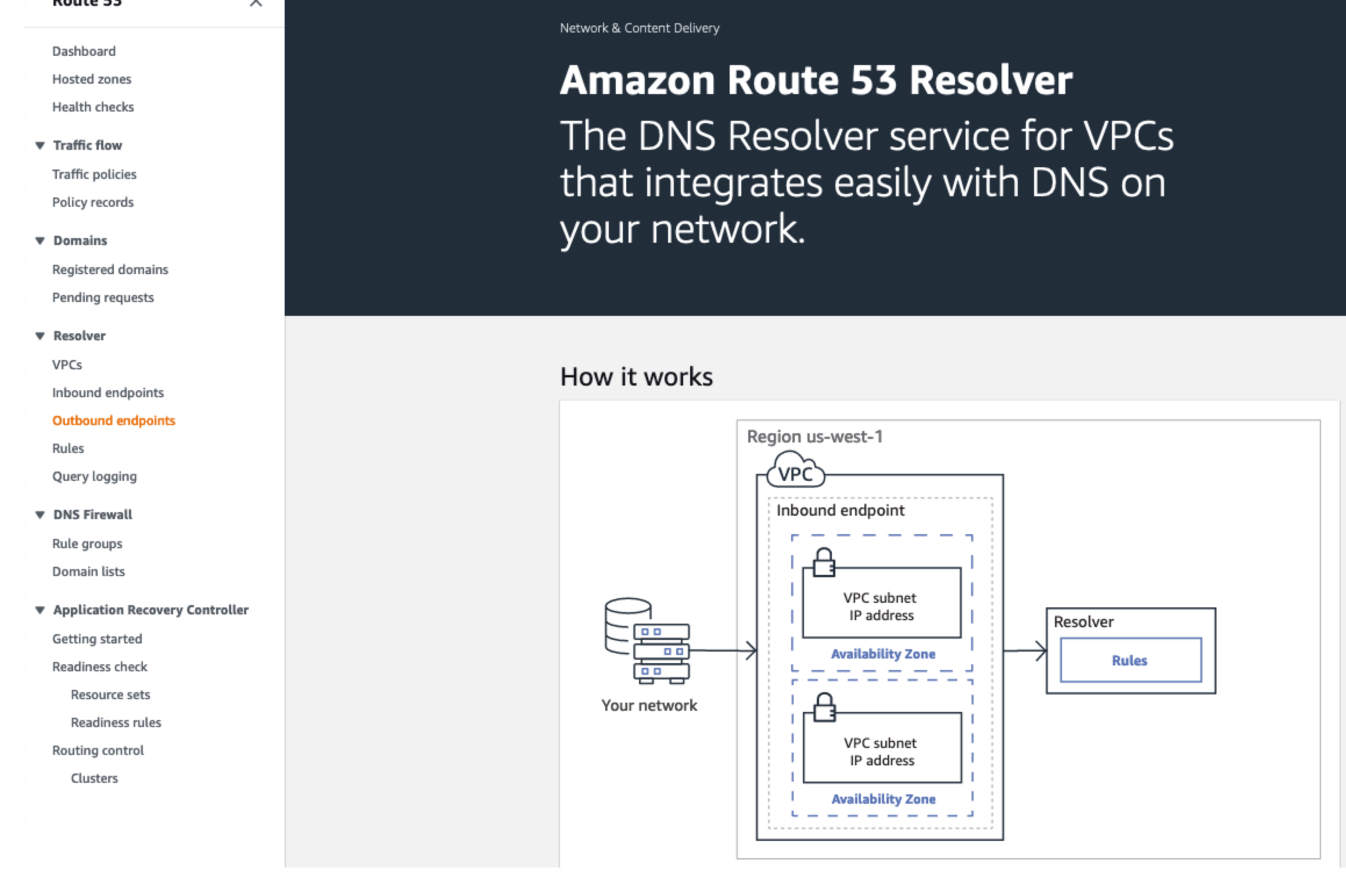
Recall that our simulated on-premises datacenter has a DNS server, providing authoritative name service for the example.corp domain where all internal application hosts are registered. In order to provide a complete hybrid-connectivity solution, we want to enable hosts in our AWS VPCs to resolve names of hosts in the datacenter environment. This can be achieved using Route 53 resolvers and conditional forwarding rules for the example.corp domain, while allowing the AWS instances to continue to take advantage of the highly available Amazon DNS service for all other name resolution inside the VPC and the internet.

For this exercise, we will focus on establishing DNS resolution from the AWS environment to the simulated datacenter, but it's important to note that the reverse is possible as well. Route 53 Resolvers supports inbound DNS queries that are conditionally forwarded from an on-premises DNS server. You can learn more about inbound DNS resolution in the [AWS documentation](#).

Configure a Route 53 Resolver Outbound Endpoint

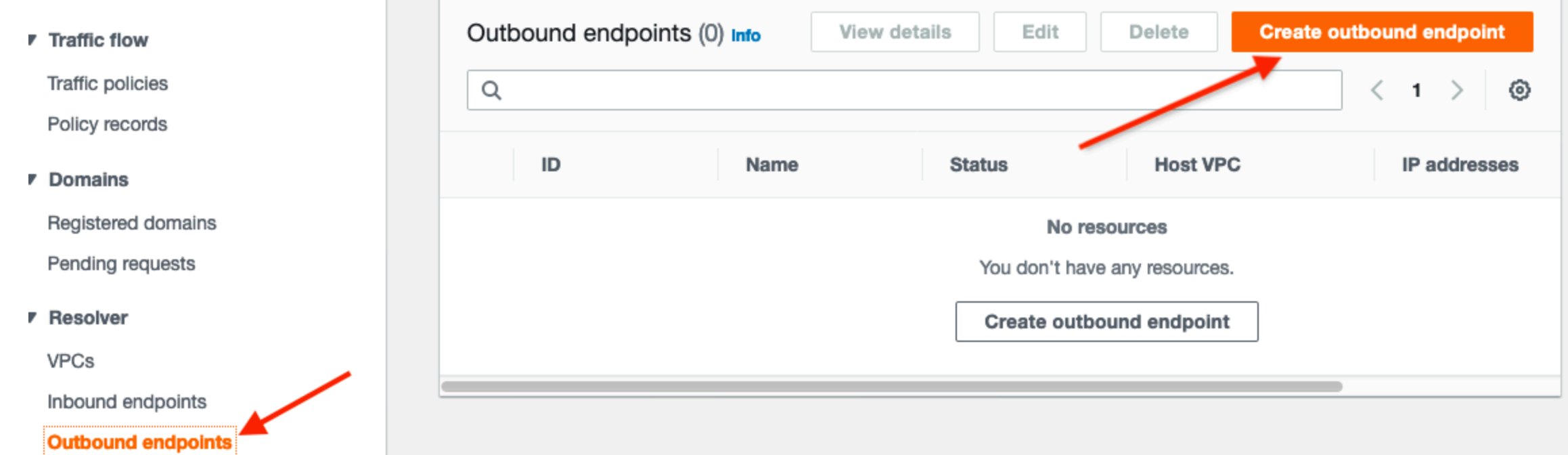
Route 53 Resolver uses endpoints to communicate with external DNS servers. An endpoint is an Elastic Network Interface (ENI) placed inside of a VPC which has connectivity to the existing DNS server. This may be a DNS server running on an EC2 instance, or a DNS server running on-premises accessible via Direct Connect or VPN. Since all three VPCs in our AWS environment have connectivity to the simulated datacenter via the Transit Gateway, we can use any of them for our endpoint. The endpoint will create interfaces in a minimum of two availability zones in your chosen VPC for high availability.

- Navigate to [Route 53](#).
- Expand the hamburger on the top left and Under **Resolver** select **Outbound endpoints**



If you have not been to the Route 53 Resolver console before, it will take you to a splash page that you see in the screenshot above. Under **Resolver** on the left menu, select **Outbound endpoints** again to take you to the screen you see in the screenshot below. You want to see the button **Create outbound endpoint** not **Configure endpoints**

- Click **Create outbound endpoint**.



- Configure the settings for the outbound endpoint as follows:

- Endpoint name:** NetworkingDayOutbound
- VPC in the Region:** Select VPC A
- Security group for this endpoint:** Select the default Security Group for your VPC A, which already contains a rule permitting outbound connectivity.
- Endpoint type:** Select IPv4
- IP address #1:**
 - Availability zone:** Choose us-east-1a
 - Subnet:** VPC A Private Subnet AZ1.
 - IP address:** Use an IP address that is selected automatically
- IP address #2:**
 - Availability zone:** us-east-1b
 - Subnet:** VPC A Private Subnet AZ2.
 - IP address:** Use an IP address that is selected automatically

Configure outbound endpoint

An outbound endpoint contains the information that Resolver needs to route DNS queries to your network from your VPCs.

General settings for outbound endpoint

Endpoint name
A friendly name lets you easily find your endpoint on the dashboard.

The endpoint name can have up to 64 characters. Valid characters: a-z, A-Z, 0-9, space, _ (underscore), and - (hyphen)

VPC in the Region: us-east-1 (N. Virginia)
All outbound DNS queries will flow through this VPC on the way from other VPCs. You can't change this value after you create the endpoint.

Security group for this endpoint
A security group controls access to this VPC. The security group that you choose must include

Endpoint Type
Route 53 Resolver endpoints support IPv4, IPv6, and Dual-stack IP addresses. For a Dual-stack connection one endpoint can use both IPv4 and IPv6 addresses to connect to a VPC.

IP addresses

To improve reliability, Resolver requires that you specify two IP addresses for DNS queries. We recommend that you specify IP addresses in two different Availability Zones. After you add the first two IP addresses, you can optionally add more in the same or different Availability Zones.

IP address #1

Availability Zone
The Availability Zone that you choose for outbound DNS queries must be configured with a subnet.

Subnet
The subnet that you choose must have an available IP address.

IPv4 address
For outbound DNS queries, you can either let the service choose an IP address for you from the available IP addresses in the subnet, or you can specify the address yourself.
☒ Use an IP address that is selected automatically
☐ Use an IP address that you specify

IP address #2

Availability Zone
The Availability Zone that you choose for outbound DNS queries must be configured with a subnet.

Subnet
The subnet that you choose must have an available IP address.

IPv4 address
For outbound DNS queries, you can either let the service choose an IP address for you from the available IP addresses in the subnet, or you can specify the address yourself.
☒ Use an IP address that is selected automatically
☐ Use an IP address that you specify

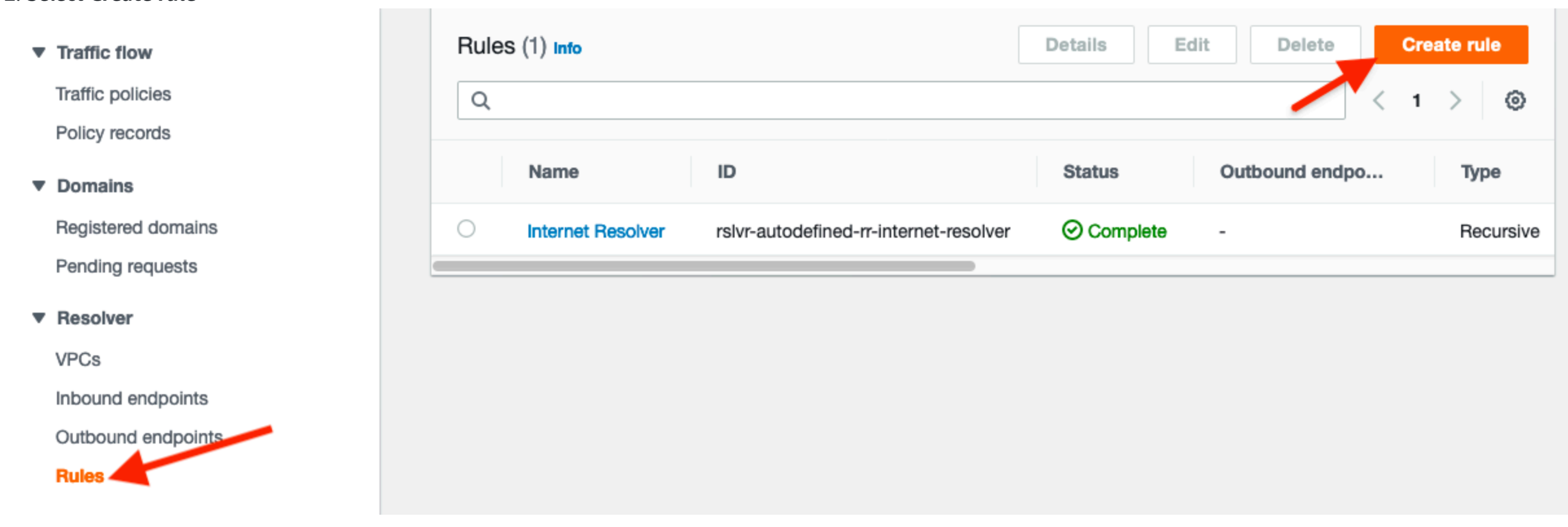
- Click **Submit** to create the endpoint.
- Wait for the endpoint status to change to **Operational**. Then proceed to the next step.

Outbound endpoints (1)					
Search					
ID	Name	Status	Host VPC	IP addresses	Rules
rsrv-out-177249b062164f689	NetworkingDayOutbound	Operational	vpc-0ba56bc7d6027311f	2	0

Create a Route 53 Resolver rule for example.corp.

Now that we have created an outbound endpoint this provides the Route 53 Resolver with a path to the on premise DNS server via VPC A and the Transit Gateway's VPN connection. Next we need to configure a *Route 53 Resolver Rule* to direct queries for **example.corp** to the on premise DNS server via that endpoint. We will associate the rule with VPCs A, B, and C and this will cause Route 53 Resolver to use this rule whenever the VPC DNS resolver processes queries for *example.corp* from instances in any of those three VPCs.

- While still in the Route 53 console, navigate to the **Rules** tab under **Resolver**
- Select **Create rule**



- Provide configuration for the rule:

- Set Name** as NetworkingDayRule
- Leave Rule type** as Forward
- Specify the **Domain name** as example.corp.
- Associate the rule with all three of your AWS VPCs: VPC A, VPC B, and VPC C.
- For **Outbound endpoint** choose the endpoint you created in the previous step: NetworkingDayOutbound
- Under **Target IP addresses** enter the on-premise DNS server IP address: 172.16.1.200

Rule for outbound traffic

For queries that originate to your VPC, you can define how to forward DNS queries out of the VPC.

Name
A friendly name helps you find your rule on the dashboard.

The rule name can have up to 64 characters. Valid characters: a-z, A-Z, 0-9, space, _ (underscore), and - (hyphen)

Rule type
Choose Forward to forward DNS queries to the IP addresses that you specify in Target IP addresses section near the bottom of this page. Choose System to have Resolver handle queries for a specified subdomain. You can't change this value after you create a rule.

Domain name
DNS queries for this domain name are forwarded to the IP address that you specify in the Target IP addresses section near the bottom of this page. If a query matches multiple rules (example.com and www.example.com), outbound DNS queries are routed using the rule that contains the most specific domain name (www.example.com). You can't change this value after you create a rule.

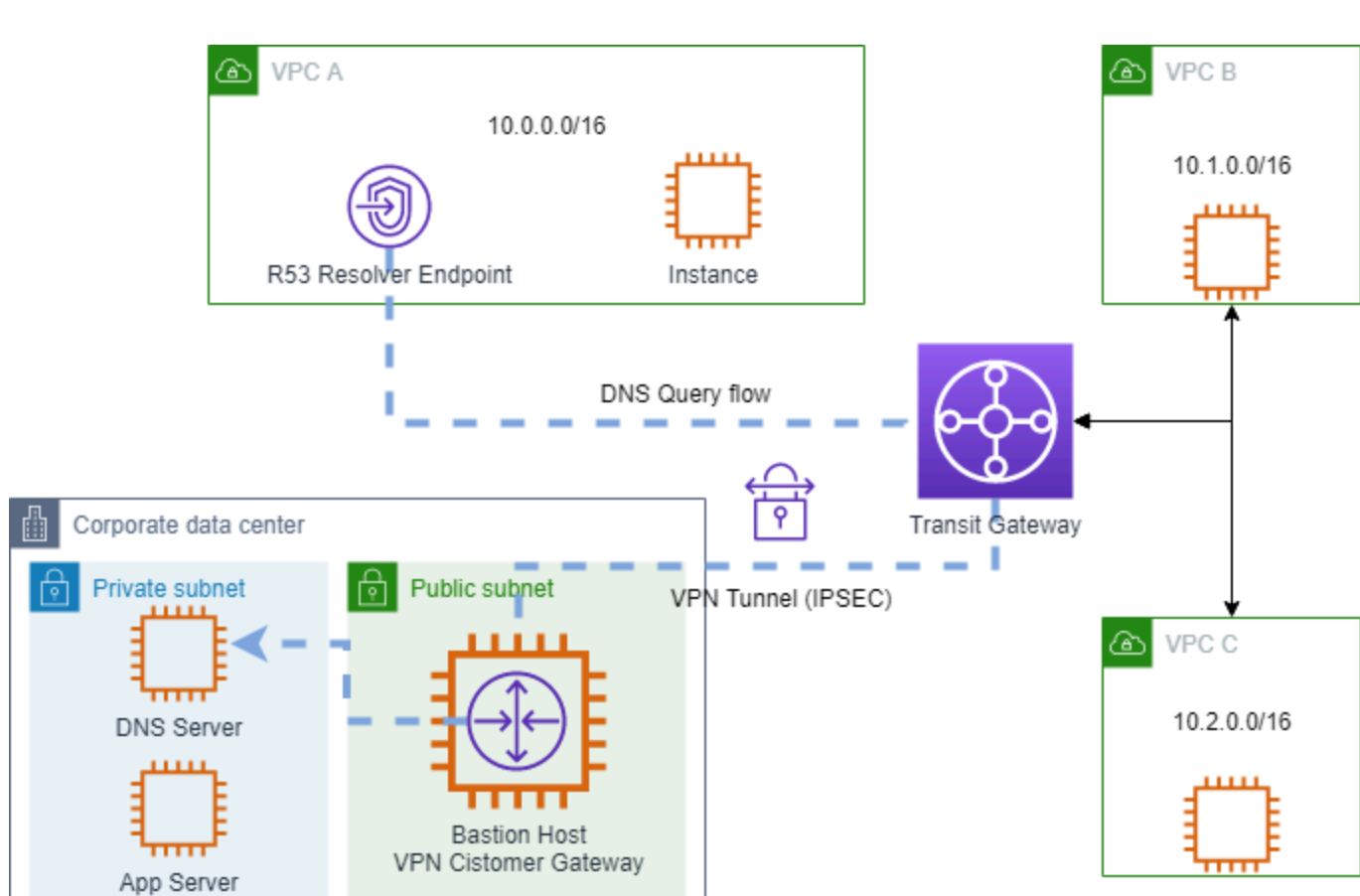
VPCs that use this rule - optional
You can associate this rule with as many VPCs as you want. To remove a VPC, choose the X for that VPC.

Outbound endpoint
Resolver uses the outbound endpoint to route DNS queries to the IP addresses that you specify in the Target IP addresses section near the bottom of this page.

Target IP addresses
DNS queries are forwarded to the following IPv4 addresses:

- Click **Submit** to create the rule.

Testing the Route 53 Resolver rule



At this point, we have configured Route 53 resolver to forward queries to the datacenter's DNS server for the example.corp domain from any of the AWS VPCs A, B, or C. We can test name resolution by attempting to connect to the on-premises app server from one of the EC2 instances in our AWS VPCs.

- In the EC2 console navigate to [Instances](#)
- Select the **check box** next to VPC B Private AZ1 Server and click the **Connect** button above.
- On the **Session manager** tab click **Connect** to open a command prompt
- Check setting of the DNS server on the instance by examining resolv.conf:

```
cat /etc/resolv.conf
```

Note that the instance is using the AWS provided DNS server (e.g. 10.1.0.2) and not your on-premises DNS server for name resolution:

```
nameserver 10.1.0.2
search ec2.internal
```

- Check that the hostname for myapp.example.corp resolves to an IP address:

```
nslookup myapp.example.corp
```

Note that the instance is using the AWS provided DNS server (e.g. 10.1.0.2) and not your on-premises DNS server for name resolution:

```
nslookup myapp.example.corp
Server:      10.1.0.2
Address:     10.1.0.2#53

Non-authoritative answer:
Name:   myapp.example.corp
Address: 172.16.1.100
```

- Use **curl** to query the on-premises application server by its hostname, myapp.example.corp

```
curl http://myapp.example.corp
```

If name resolution and your application server are functioning, you will receive back a "Hello, world" response.

```
curl http://myapp.example.corp
Hello, world.
```

- Terminate the Session Manager connection.

Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept

Decline

Customize